Flash Attention on FPGA

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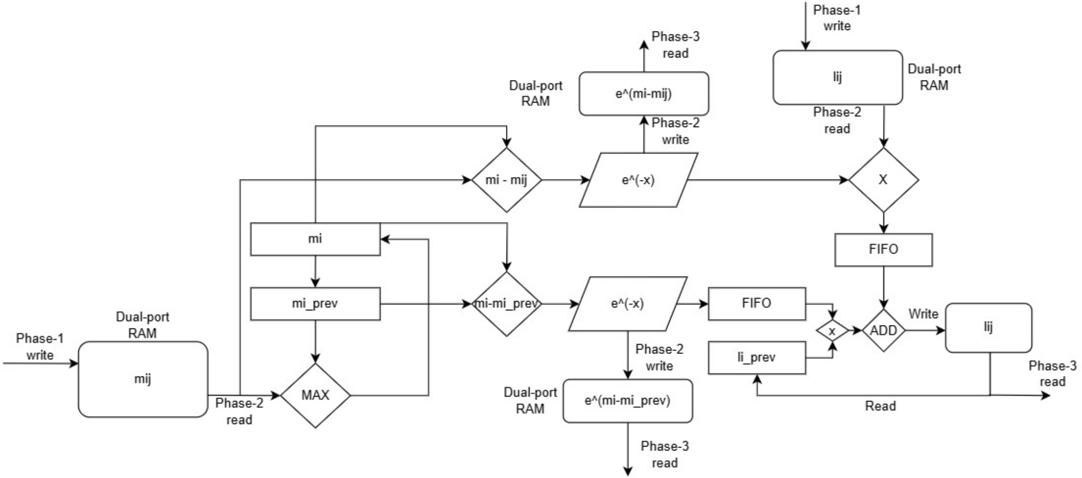
Overview

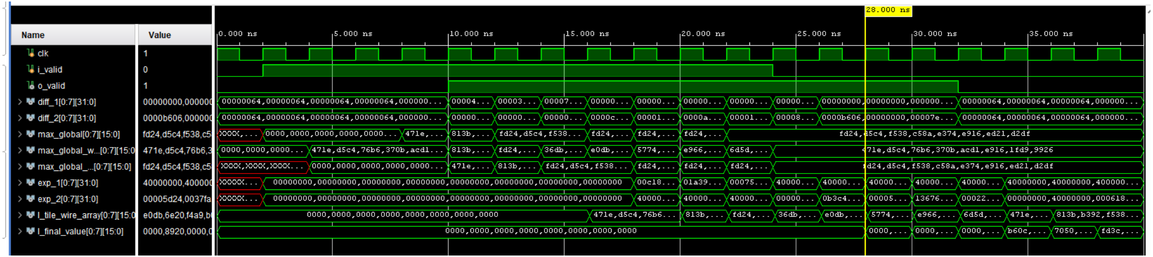
1. Phase-2 Architecture & Performance
2. Phase-3
3. Future Work

Phase-2 Hardware Architecture



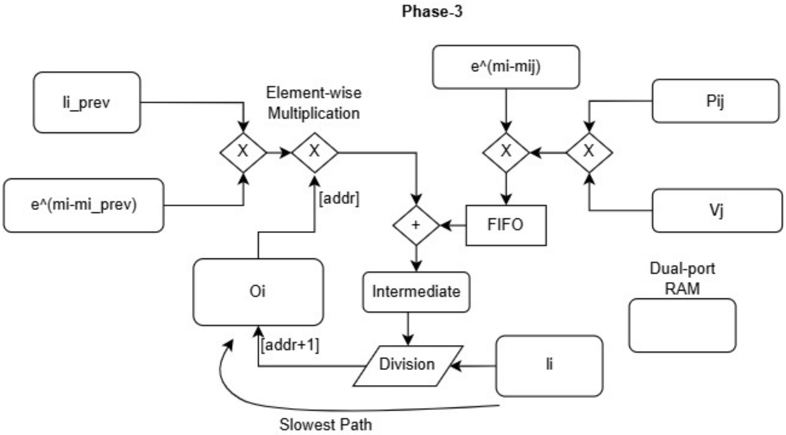
Items to look out of for: **Dual-port RAM, FIFO**

Phase-2 Performance



- BRAM access: 2 CC, comparison and storage: 2 CC, subtraction: 1 CC, e^{-x}: 3 CC, Pipelined Multiplication: 3 CC and addition: 1 CC. Total = 13 CC latency
- Giving us an initial latency of 80 ns and 12 ns for every other output after that. The final formula for the total latency for a tile is $\{80 + 12(n-1)\text{ ns}\}$, where $n = \left\lfloor \frac{B_r}{W} \right\rfloor + 1$ (Calculations done for 162 MHz)
- General formula: Latency $= \frac{13}{f_{op}} + \frac{2(n-1)}{f_{op}}$
- Further improvements: Improve BRAM access, and Multiplier's efficiency - dependent on Alveo V80.

Phase-3 Architecture



THINGS TO NOTE: Large number of multiplications, on-chip memory: BRAM/URAM, Divison algorithm on average has 25 CC latency for 16-bit accuracy, FIFO to compensate for the delay due to the division

Need for faster Division algorithm

1. The output latency per tile is **directly proportional** to the latency of the division algorithm
2. Ways of optimization: Use **8-bit** approximations for division, use LUT-based implementation to use fewer DSP slices, division IP (still suffers from the latency issue at some level)

Currently referring papers for **Low-Latency** implementation of the Division algorithm

Future Work

1. Complete RTL implementation of Phase-3 with integration with Phase-2 with implementation.
2. Verification of Phase-2 on the Alveo V80 FPGA.
3. More info on AVED configuration on V80.

THE END

Feedback & Improvement Ideas?